Intensity, decibels and Doppler

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish intensity change from frequency shift.

SUCCESS CRITERIA - I can...

1. Use an intensity ratio separately from pitch.
2. Interpret a logarithmic sound-level difference.
3. Predict approach and recession Doppler trends.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “intensity” and “decibel” to compare your model or result. Sentence starters are optional.

intensity: power passing through a unit area

decibel: a unit for a logarithmic sound-level comparison

Doppler shift: a frequency change associated with relative motion

01 Bring back an earlier idea

A 500 Hz sound travels through air. Do air particles mainly move parallel or perpendicular to propagation?
Name the wave type.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta\beta = 10 \log_{10}(I_2/I_1)$.
I is intensity; β is sound level in dB. Tenfold intensity gives +10 dB. Approach generally raises observed frequency; recession lowers it.

Worked reasoning: For the same point source, doubling distance spreads its power over four times the area, so ideal intensity drops to one quarter. This assumes spherical spreading without extra absorption.

Your turn: Draw two concentric wavefronts and indicate why the outer surface shares the power more widely.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Look and explain

A moving source approaches a stationary listener, then recedes.
Sketch closer wavefront spacing in front and wider spacing behind; compare heard pitch.

Sentence starter: Both … . They differ in … because … .

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

04 Build the solution

A supplied rule says 10 times intensity adds 10 dB.
Compare level changes for 10 times and 100 times intensity.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A loud stationary siren and a quiet approaching siren are compared. Does “louder” alone mean “higher frequency”? **Explain**.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A sample warms by 4°C after a certain energy input. If absorbed energy becomes 6 times as large with the same mass and material and no phase change, find the new rise.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

An observer hears a steadily emitting siren approach. Is observed frequency higher or lower than emitted frequency?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Compare intensities at 1, 2 and 3 m from the same ideal source as fractions of the 1 m value.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3